Experiment 1 Hypothesis Tests

Packages

```
required_packages <- c(
  "dplyr",
  "tidyr",
  "ggplot2",
  "jsonlite",
  "knitr",
  "lme4",
  "lmerTest",
  "emmeans"
)

missing_packages <- required_packages[!vapply(required_packages, requireNamespace, logical(1), quietly
= TRUE)]
if (length(missing_packages) > 0) {
  stop(
    "Missing required packages: ",
    paste(missing_packages, collapse = ", "),
    ". Install them before rendering."
  )
}

lapply(required_packages, library, character.only = TRUE)
```

```
## [[1]]
## [1] "dplyr"     "stats"     "graphics"  "grDevices" "utils"     "datasets"
## [7] "methods"   "base"
##
## [[2]]
## [1] "tidyr"     "dplyr"     "stats"     "graphics"  "grDevices" "utils"
## [7] "datasets"  "methods"   "base"
##
## [[3]]
##  [1] "ggplot2"   "tidyr"     "dplyr"     "stats"     "graphics"  "grDevices"
##  [7] "utils"     "datasets"  "methods"   "base"
##
## [[4]]
##  [1] "jsonlite"  "ggplot2"   "tidyr"     "dplyr"     "stats"     "graphics"
##  [7] "grDevices" "utils"     "datasets"  "methods"   "base"
##
## [[5]]
##  [1] "knitr"     "jsonlite"  "ggplot2"   "tidyr"     "dplyr"     "stats"
##  [7] "graphics"  "grDevices" "utils"     "datasets"  "methods"   "base"
##
## [[6]]
##  [1] "lme4"      "Matrix"    "knitr"     "jsonlite"  "ggplot2"   "tidyr"
##  [7] "dplyr"     "stats"     "graphics"  "grDevices" "utils"     "datasets"
## [13] "methods"   "base"
##
## [[7]]
##  [1] "lmerTest"  "lme4"      "Matrix"    "knitr"     "jsonlite"  "ggplot2"
##  [7] "tidyr"     "dplyr"     "stats"     "graphics"  "grDevices" "utils"
## [13] "datasets"  "methods"   "base"
##
## [[8]]
##  [1] "emmeans"   "lmerTest"  "lme4"      "Matrix"    "knitr"     "jsonlite"
##  [7] "ggplot2"   "tidyr"     "dplyr"     "stats"     "graphics"  "grDevices"
## [13] "utils"     "datasets"  "methods"   "base"
```

Paths

```
report_dir <- local({
  current_input <- tryCatch(
    knitr::current_input(),
    error = function(e) character(0)
  )

  if (length(current_input) == 1 && nzchar(current_input)) {
    input_path <- current_input
    if (!grepl("^(/|[A-Za-z]:[/\\\\])", input_path)) {
      input_path <- file.path(getwd(), input_path)
    }

    if (grepl("\\.[Rr]md$", input_path) || file.exists(input_path)) {
      return(normalizePath(dirname(input_path), winslash = "/", mustWork = TRUE))
    }
  }

  fallback_dir <- file.path(getwd(), "Experiment1", "analysis")
  normalizePath(fallback_dir, winslash = "/", mustWork = TRUE)
})

resolve_report_path <- function(path_value) {
  if (grepl("^(/|[A-Za-z]:[/\\\\])", path_value)) {
    return(normalizePath(path_value, winslash = "/", mustWork = TRUE))
  }

  normalizePath(
    file.path(report_dir, path_value),
    winslash = "/",
    mustWork = TRUE
  )
}

data_dir <- resolve_report_path(params$data_dir)
```

Helpers

```
TREATMENT_CONDITIONS <- c(2, 3, 4, 5, 6, 9, 18, 19, 20)

TRUST_ITEM_KEYS <- c(
  "skeptical_rating",
  "data_trust",
  "usability_difficulty",
  "comprehension_ease"
)

INTERACTION_ITEM_KEYS <- c(
  "navigation_control",
  "content_control",
  "pace_control",
  "interface_exploration",
  "interface_responsiveness",
  "user_communication",
  "personal_conversation",
  "interface_interaction",
  "interface_sensitivity"
)

extract_condition_number <- function(condition_id) {
  if (is.na(condition_id) || !nzchar(condition_id)) {
    return(NA_real_)
  }

  parts <- strsplit(as.character(condition_id), "_")[[1]]
  hits <- parts[grepl("^[0-9]+$", parts)]

  if (length(hits) == 0) {
    return(NA_real_)
  }

  as.numeric(hits[[1]])
}

load_participant_data <- function(file_path) {
  df <- read.csv(file_path, stringsAsFactors = FALSE)
  df$source_file <- basename(file_path)

  if (!"participant_id" %in% names(df) || all(is.na(df$participant_id))) {
    participant_id <- strsplit(
      tools::file_path_sans_ext(basename(file_path)),
      "_"
    )[[1]][2]
    df$participant_id <- participant_id
  }

  if ("condition_id" %in% names(df)) {
    df$condition_id[is.na(df$condition_id)] <- "unknown"
    df$condition_number <- vapply(
      df$condition_id,
      extract_condition_number,
      numeric(1)
    )
  } else {
    df$condition_id <- "unknown"
    df$condition_number <- NA_real_
```

```
  }

  df
}

parse_response_json <- function(value) {
  if (is.na(value) || !nzchar(value)) {
    return(list())
  }

  parsed <- tryCatch(
    jsonlite::fromJSON(value, simplifyVector = TRUE),
    error = function(e) list()
  )

  if (is.null(parsed) || length(parsed) == 0) {
    return(list())
  }

  parsed
}

expand_response_columns <- function(data) {
  parsed <- lapply(data$response, parse_response_json)
  expanded <- bind_rows(
    lapply(parsed, function(x) {
      if (length(x) == 0) {
        return(data.frame())
      }
      as.data.frame(x, stringsAsFactors = FALSE)
    })
  )

  if (nrow(expanded) == 0) {
    expanded <- data.frame(row.names = seq_len(nrow(data)))
  }

  if (nrow(expanded) < nrow(data)) {
    expanded[(nrow(expanded) + 1):nrow(data), ] <- NA
  }

  overlap <- intersect(names(expanded), names(data))
  if (length(overlap) > 0) {
    data <- data[, setdiff(names(data), overlap), drop = FALSE]
  }

  bind_cols(data, expanded)
}

normalize_likert_0_6_to_1_7 <- function(values) {
  numeric_values <- suppressWarnings(as.numeric(values))
  valid <- numeric_values[!is.na(numeric_values)]

  if (length(valid) == 0) {
    return(numeric_values)
  }

  if (all(valid >= 0 & valid <= 6)) {
    return(pmin(pmax(numeric_values + 1, 1), 7))
  }
```

```
  numeric_values
}

reverse_likert_1_7 <- function(values) {
  8 - suppressWarnings(as.numeric(values))
}

map_interaction_group <- function(condition_number) {
  dplyr::case_when(
    condition_number %in% c(2, 3, 9) ~ "Static",
    condition_number %in% c(4, 5, 6) ~ "Interactive",
    condition_number %in% c(18, 19, 20) ~ "Buggy",
    TRUE ~ NA_character_
  )
}

map_vis_type <- function(condition_number) {
  dplyr::case_when(
    condition_number %in% c(2, 5, 19) ~ "PI",
    condition_number %in% c(3, 4, 18) ~ "Ensemble",
    condition_number %in% c(6, 9, 20) ~ "PI + Ensemble",
    TRUE ~ NA_character_
  )
}

safe_min <- function(x) {
  x <- suppressWarnings(as.numeric(x))
  x <- x[is.finite(x)]
  if (length(x) == 0) {
    return(NA_real_)
  }
  min(x)
}

safe_max <- function(x) {
  x <- suppressWarnings(as.numeric(x))
  x <- x[is.finite(x)]
  if (length(x) == 0) {
    return(NA_real_)
  }
  max(x)
}

set_sum_contrasts <- function(data, factors) {
  out <- data
  for (factor_name in factors) {
    out[[factor_name]] <- factor(out[[factor_name]])
    contrasts(out[[factor_name]]) <- contr.sum(length(levels(out[[factor_name]])))
  }
  out
}

augment_contrast_table <- function(table_df, family_label, alpha, adjust_method) {
  out <- as.data.frame(table_df)
  out$family <- family_label
  out$p_adjusted <- p.adjust(out$p.value, method = adjust_method)
  out$alpha <- alpha
  out
}
```

Load and Prepare Data

```
csv_files <- sort(list.files(
  data_dir,
  pattern = params$csv_pattern,
  full.names = TRUE
))

raw_data <- bind_rows(lapply(csv_files, load_participant_data)) %>%
  filter(!is.na(participant_id)) %>%
  filter(tolower(as.character(participant_id)) != "test")

prediction_rows_raw <- raw_data %>%
  filter(trial_type == "prediction-task")

trust_rows_raw <- raw_data %>%
  filter(trial_type == "trust-survey")

trust_rows_expanded <- expand_response_columns(trust_rows_raw)

for (col in c(TRUST_ITEM_KEYS, INTERACTION_ITEM_KEYS)) {
  if (col %in% names(trust_rows_expanded)) {
    trust_rows_expanded[[col]] <- normalize_likert_0_6_to_1_7(trust_rows_expanded[[col]])
  }
}

trust_treatment <- trust_rows_expanded %>%
  mutate(
    condition_number = suppressWarnings(as.numeric(condition_number)),
    vis_type = map_vis_type(condition_number),
    interaction_group = map_interaction_group(condition_number),
    participant_id = as.character(participant_id)
  ) %>%
  filter(condition_number %in% TREATMENT_CONDITIONS) %>%
  filter(!is.na(vis_type), !is.na(interaction_group))

for (col in c(TRUST_ITEM_KEYS, INTERACTION_ITEM_KEYS)) {
  if (col %in% names(trust_treatment)) {
    trust_treatment[[col]] <- suppressWarnings(as.numeric(trust_treatment[[col]]))
  }
}

trust_treatment <- trust_treatment %>%
  mutate(
    skeptical_rating_scored = reverse_likert_1_7(skeptical_rating),
    usability_difficulty_scored = reverse_likert_1_7(usability_difficulty),
    trust_composite = rowMeans(
      cbind(
        skeptical_rating_scored,
        data_trust,
        usability_difficulty_scored,
        comprehension_ease
      ),
      na.rm = TRUE
    ),
    interaction_composite = rowMeans(
      dplyr::select(., dplyr::all_of(INTERACTION_ITEM_KEYS)),
      na.rm = TRUE
    )
  )
```

```
trust_treatment$trust_composite[!is.finite(trust_treatment$trust_composite)] <- NA_real_
trust_treatment$interaction_composite[!is.finite(trust_treatment$interaction_composite)] <- NA_real_

complete_participants <- trust_treatment %>%
  group_by(participant_id) %>%
  summarise(
    n_interaction_groups = n_distinct(interaction_group),
    n_treatment_rows = n(),
    .groups = "drop"
  ) %>%
  filter(n_interaction_groups == 3, n_treatment_rows >= 3) %>%
  pull(participant_id)

h2_data <- trust_treatment %>%
  filter(participant_id %in% complete_participants) %>%
  filter(is.finite(trust_composite)) %>%
  mutate(
    participant_id = factor(participant_id),
    vis_type = factor(vis_type, levels = c("PI", "Ensemble", "PI + Ensemble")),
    interaction_group = factor(interaction_group, levels = c("Static", "Interactive", "Buggy"))
  ) %>%
  droplevels()

h3_data <- trust_treatment %>%
  filter(participant_id %in% complete_participants) %>%
  filter(is.finite(interaction_composite)) %>%
  mutate(
    participant_id = factor(participant_id),
    vis_type = factor(vis_type, levels = c("PI", "Ensemble", "PI + Ensemble")),
    interaction_group = factor(interaction_group, levels = c("Static", "Interactive", "Buggy"))
  ) %>%
  droplevels()

prediction_rows_prepared <- prediction_rows_raw %>%
  mutate(
    phase = suppressWarnings(as.numeric(phase)),
    condition_number = suppressWarnings(as.numeric(condition_number)),
    vis_type = map_vis_type(condition_number),
    interaction_group = map_interaction_group(condition_number),
    confidence_rating = suppressWarnings(as.numeric(confidence_rating)),
    participant_id = as.character(participant_id)
  ) %>%
  filter(is.finite(confidence_rating))

h1_baseline <- prediction_rows_prepared %>%
  filter(phase == 1) %>%
  group_by(participant_id) %>%
  summarise(
    baseline_confidence = mean(confidence_rating, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  filter(is.finite(baseline_confidence))

h1_phase2_condition <- prediction_rows_prepared %>%
  filter(phase %in% c(2, 3, 4)) %>%
  filter(condition_number %in% TREATMENT_CONDITIONS) %>%
  filter(!is.na(vis_type), !is.na(interaction_group), is.finite(condition_number)) %>%
  group_by(participant_id, condition_number, condition_name, vis_type, interaction_group) %>%
  summarise(
    confidence_rating = mean(confidence_rating, na.rm = TRUE),
```

```
    .groups = "drop"
  ) %>%
  filter(is.finite(confidence_rating))

h1_data <- h1_phase2_condition %>%
  inner_join(h1_baseline, by = "participant_id") %>%
  mutate(
    confidence_shift = confidence_rating - baseline_confidence,
    participant_id = factor(participant_id),
    vis_type = factor(vis_type, levels = c("PI", "Ensemble", "PI + Ensemble")),
    interaction_group = factor(interaction_group, levels = c("Static", "Interactive", "Buggy"))
  ) %>%
  filter(is.finite(confidence_shift)) %>%
  droplevels()

h1_summary <- h1_data %>%
  group_by(vis_type) %>%
  summarise(
    n = n_distinct(participant_id),
    mean_shift = mean(confidence_shift, na.rm = TRUE),
    sd_shift = sd(confidence_shift, na.rm = TRUE),
    se_shift = sd_shift / sqrt(pmax(n, 1)),
    .groups = "drop"
  ) %>%
  mutate(
    group_category = factor(vis_type, levels = c("PI", "Ensemble", "PI + Ensemble"), ordered = TRUE)
  ) %>%
  arrange(group_category)

h1_data <- h1_data %>%
  mutate(
    vis_type = factor(vis_type, levels = c("PI", "Ensemble", "PI + Ensemble"))
  ) %>%
  droplevels()
```

Sample Flow and Cell Counts

```
sample_flow <- data.frame(
  stage = c(
    "Participant CSV files",
    "Participants after dropping test IDs",
    "Prediction rows (all)",
    "Trust-survey rows (all)",
    "Participants with complete treatment rows (Static + Interactive + Buggy)",
    "H1 rows used",
    "H2 rows used",
    "H3 rows used"
  ),
  value = c(
    length(csv_files),
    dplyr::n_distinct(raw_data$participant_id),
    nrow(prediction_rows_raw),
    nrow(trust_rows_raw),
    length(complete_participants),
    nrow(h1_data),
    nrow(h2_data),
    nrow(h3_data)
  )
)

knitr::kable(sample_flow, align = c("l", "r"))
```

stage	value
Participant CSV files	337
Participants after dropping test IDs	337
Prediction rows (all)	1340
Trust-survey rows (all)	1340
Participants with complete treatment rows (Static + Interactive + Buggy)	335
H1 rows used	1005
H2 rows used	1005
H3 rows used	1005

```
h1_cell_counts <- h1_data %>%
  count(vis_type, name = "n")

h2_cell_counts <- h2_data %>%
  count(interaction_group, vis_type, name = "n")

knitr::kable(h1_cell_counts, caption = "H1 Cell Counts: Confidence Shift Rows by Visualization Family")
```

H1 Cell Counts: Confidence Shift Rows by Visualization Family

vis_type	n
PI	340
Ensemble	330
PI + Ensemble	335

```
knitr::kable(h2_cell_counts, caption = "H2/H3 Cell Counts: Trust-Survey Rows by Interaction Group and V
isualization Family")
```

H2/H3 Cell Counts: Trust-Survey Rows by Interaction Group and Visualization Family

interaction_group	vis_type	n
Static	PI	114
Static	Ensemble	109
Static	PI + Ensemble	112
Interactive	PI	117
Interactive	Ensemble	109
Interactive	PI + Ensemble	109
Buggy	PI	109
Buggy	Ensemble	112
Buggy	PI + Ensemble	114

Preprocessing Validation

```
range_check <- bind_rows(
  data.frame(
    scale = "Trust composite items (normalized)",
    item = TRUST_ITEM_KEYS,
    min_value = vapply(TRUST_ITEM_KEYS, function(k) safe_min(trust_treatment[[k]]), numeric(1)),
    max_value = vapply(TRUST_ITEM_KEYS, function(k) safe_max(trust_treatment[[k]]), numeric(1))
  ),
  data.frame(
    scale = "Interaction composite items (normalized)",
    item = INTERACTION_ITEM_KEYS,
    min_value = vapply(INTERACTION_ITEM_KEYS, function(k) safe_min(trust_treatment[[k]]), numeric(1)),
    max_value = vapply(INTERACTION_ITEM_KEYS, function(k) safe_max(trust_treatment[[k]]), numeric(1))
  )
)

condition_mapping_check <- trust_treatment %>%
  distinct(condition_number, condition_name, interaction_group, vis_type) %>%
  arrange(condition_number)

knitr::kable(range_check, digits = 2, caption = "Likert Range Checks After 0-6 -> 1-7 Normalization")
```

Likert Range Checks After 0-6 -> 1-7 Normalization

	scale	item	min_value	max_value
skeptical_rating	Trust composite items (normalized)	skeptical_rating	1	7
data_trust	Trust composite items (normalized)	data_trust	1	7
usability_difficulty	Trust composite items (normalized)	usability_difficulty	1	7
comprehension_ease	Trust composite items (normalized)	comprehension_ease	1	7
navigation_control	Interaction composite items (normalized)	navigation_control	1	7
content_control	Interaction composite items (normalized)	content_control	1	7
pace_control	Interaction composite items (normalized)	pace_control	1	7
interface_exploration	Interaction composite items (normalized)	interface_exploration	1	7
interface_responsiveness	Interaction composite items (normalized)	interface_responsiveness	1	7
user_communication	Interaction composite items (normalized)	user_communication	1	7

	scale	item	min_value	max_value
personal_conversation	Interaction composite items (normalized)	personal_conversation	1	7
interface_interaction	Interaction composite items (normalized)	interface_interaction	1	7
interface_sensitivity	Interaction composite items (normalized)	interface_sensitivity	1	7

```
knitr::kable(condition_mapping_check, caption = "Condition Mapping Check")
```

Condition Mapping Check

condition_number	condition_name	interaction_group	vis_type
2	PI Plot	Static	PI
3	Ensemble Plot	Static	Ensemble
4	Ensemble + Hover	Interactive	Ensemble
5	PI Plot + Hover	Interactive	PI
6	PI to Ensemble	Interactive	PI + Ensemble
9	Combined PI + Ensemble	Static	PI + Ensemble
18	Glitch Ensemble + Hover	Buggy	Ensemble
19	Glitch PI Plot + Hover	Buggy	PI
20	Glitch PI to Ensemble	Buggy	PI + Ensemble

H1: Confidence Shift from Baseline by Visualization Family

```
h1_participant_means <- h1_data %>%
  group_by(participant_id, vis_type) %>%
  summarise(
    baseline_confidence = mean(baseline_confidence, na.rm = TRUE),
    treatment_confidence = mean(confidence_rating, na.rm = TRUE),
    confidence_shift = mean(confidence_shift, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  filter(
    is.finite(baseline_confidence),
    is.finite(treatment_confidence),
    is.finite(confidence_shift)
  )

h1_anova_data <- bind_rows(
  h1_participant_means %>%
    distinct(participant_id, baseline_confidence) %>%
    transmute(
      participant_id,
      vis_type = "Baseline",
      confidence = baseline_confidence
    ),
  h1_participant_means %>%
    transmute(
      participant_id,
      vis_type = as.character(vis_type),
      confidence = treatment_confidence
    )
) %>%
  mutate(
    participant_id = factor(participant_id),
    vis_type = factor(vis_type, levels = c("Baseline", "PI", "Ensemble", "PI + Ensemble"))
  ) %>%
  filter(is.finite(confidence))

h1_model_data <- set_sum_contrasts(h1_anova_data, factors = c("vis_type"))

h1_anova_model <- lm(
  confidence ~ vis_type + participant_id,
  data = h1_model_data
)

h1_anova <- anova(h1_anova_model)

h1_emmeans <- as.data.frame(
  summary(emmeans(h1_anova_model, ~ vis_type), infer = FALSE)
)

h1_singular <- NA

knitr::kable(h1_summary, digits = 4, caption = "H1 Summary: Confidence Shift from Baseline by Visualiza
tion Family")
```

H1 Summary: Confidence Shift from Baseline by Visualization Family

vis_type	n	mean_shift	sd_shift	se_shift	group_category
PI	226	-0.2147	1.0799	0.0718	PI
Ensemble	221	-0.3061	1.3948	0.0938	Ensemble

vis_type	n	mean_shift	sd_shift	se_shift	group_category
PI + Ensemble	335	-0.4418	1.3499	0.0738	PI + Ensemble

```
h1_anova
```

```
## Analysis of Variance Table
##
## Response: confidence
##                 Df  Sum Sq Mean Sq F value    Pr(>F)
## vis_type         3   36.58 12.1940  15.901  4.83e-10 ***
## participant_id 334 1514.56  4.5346   5.913 < 2.2e-16 ***
## Residuals      779  597.40  0.7669
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

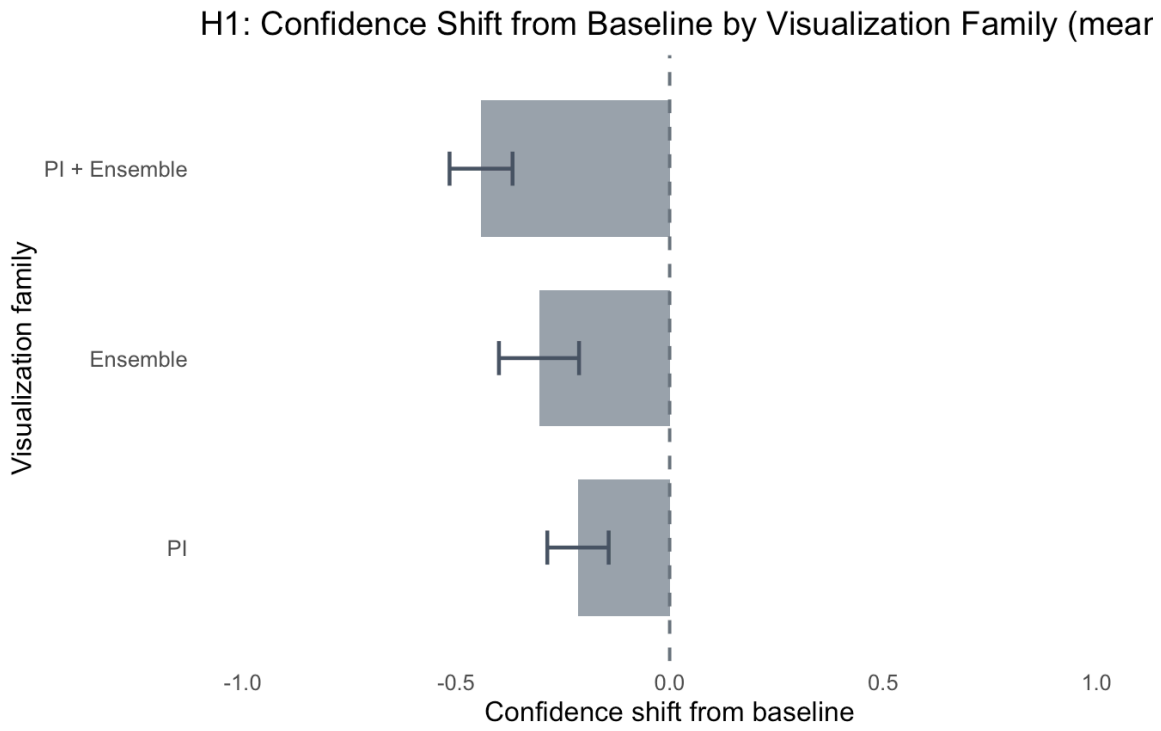
```
knitr::kable(
  h1_emmeans %>%
    transmute(
      vis_type,
      mean = emmean,
      se = SE
    ),
  digits = 4,
  caption = "H1 Estimated Marginal Means by Visualization Family (including Baseline)"
)
```

H1 Estimated Marginal Means by Visualization Family (including Baseline)

vis_type	mean	se
Baseline	4.4000	0.0478
PI	4.1882	0.0617
Ensemble	4.1089	0.0625
PI + Ensemble	3.9582	0.0478

```
h1_plot_data <- h1_summary %>%
  mutate(vis_type = factor(vis_type, levels = c("PI", "Ensemble", "PI + Ensemble"), ordered = TRUE))

ggplot(h1_plot_data, aes(y = vis_type, x = mean_shift)) +
  geom_col(fill = "#9CA3AF", width = 0.72) +
  geom_errorbarh(
    aes(xmin = mean_shift - se_shift, xmax = mean_shift + se_shift),
    height = 0.18,
    color = "#4B5563",
    linewidth = 0.8
  ) +
  geom_vline(xintercept = 0, color = "#6c757d", linetype = "dashed", linewidth = 0.7) +
  scale_x_continuous(limits = c(-1, 1)) +
  labs(
    title = "H1: Confidence Shift from Baseline by Visualization Family (mean +/- SE)",
    x = "Confidence shift from baseline",
    y = "Visualization family"
  ) +
  theme_minimal(base_size = 12) +
  theme(panel.grid = element_blank())
```



H2: Trust Composite and Interaction Availability by

Visualization Family

```
h2_model_data <- set_sum_contrasts(h2_data, factors = c("interaction_group", "vis_type"))

h2_model <- lm(
  trust_composite ~ interaction_group * vis_type + participant_id,
  data = h2_model_data,
  na.action = na.exclude
)

h2_singular <- NA

h2_condition_means <- as.data.frame(
  summary(emmeans(h2_model, ~ interaction_group * vis_type), infer = FALSE)
) %>%
  mutate(
    condition = paste(interaction_group, vis_type)
  )

h2_interaction_means <- as.data.frame(
  summary(emmeans(h2_model, ~ interaction_group), infer = FALSE)
)

h2_vis_means <- as.data.frame(
  summary(emmeans(h2_model, ~ vis_type), infer = FALSE)
)

knitr::kable(
  h2_condition_means %>%
    transmute(
      condition,
      interaction_group,
      vis_type,
      mean = emmean,
      se = SE
    ),
  digits = 4,
  caption = "H2 Estimated Means by Individual Condition (Interaction x Visualization)"
)
```

H2 Estimated Means by Individual Condition (Interaction x Visualization)

condition	interaction_group	vis_type	mean	se
Static PI	Static	PI	4.9118	0.0950
Interactive PI	Interactive	PI	4.9671	0.0937
Buggy PI	Buggy	PI	4.5950	0.0973
Static Ensemble	Static	Ensemble	4.0459	0.0973
Interactive Ensemble	Interactive	Ensemble	4.6800	0.0973
Buggy Ensemble	Buggy	Ensemble	4.3967	0.0959
Static PI + Ensemble	Static	PI + Ensemble	4.0496	0.0959
Interactive PI + Ensemble	Interactive	PI + Ensemble	4.5571	0.0973
Buggy PI + Ensemble	Buggy	PI + Ensemble	4.2650	0.0950

```
knitr::kable(
  h2_interaction_means %>%
    transmute(
      interaction_group,
      mean = emmean,
      se = SE
    ),
  digits = 4,
  caption = "H2 Estimated Means by Interaction Group"
)
```

H2 Estimated Means by Interaction Group

interaction_group	mean	se
Static	4.3358	0.0462
Interactive	4.7347	0.0462
Buggy	4.4189	0.0462

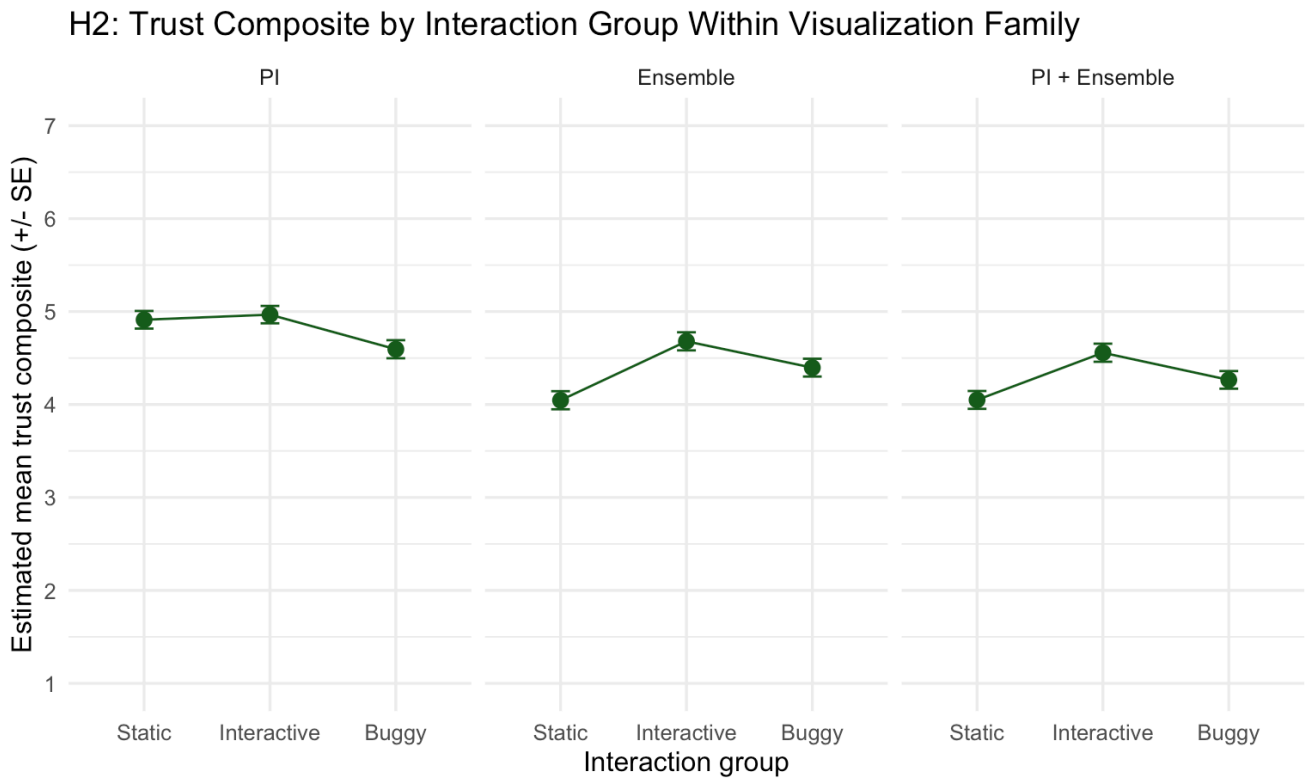
```
knitr::kable(
  h2_vis_means %>%
    transmute(
      vis_type,
      mean = emmean,
      se = SE
    ),
  digits = 4,
  caption = "H2 Estimated Means by Visualization Family"
)
```

H2 Estimated Means by Visualization Family

vis_type	mean	se
PI	4.8246	0.0549
Ensemble	4.3742	0.0560
PI + Ensemble	4.2906	0.0462

```
h2_plot_data <- h2_condition_means %>%
  mutate(
    ymin = pmax(1, emmean - SE),
    ymax = pmin(7, emmean + SE)
  )

ggplot(h2_plot_data, aes(x = interaction_group, y = emmean, group = 1)) +
  geom_point(size = 2.8, color = "#1B5E20") +
  geom_line(color = "#1B5E20") +
  geom_errorbar(aes(ymin = ymin, ymax = ymax), width = 0.15, color = "#1B5E20") +
  facet_wrap(~ vis_type, nrow = 1) +
  scale_y_continuous(limits = c(1, 7), breaks = 1:7) +
  labs(
    title = "H2: Trust Composite by Interaction Group Within Visualization Family",
    x = "Interaction group",
    y = "Estimated mean trust composite (+/- SE)"
  ) +
  theme_minimal(base_size = 12)
```



H3: Interaction Experience Composite Across Interaction

Availability

```
h3_model_data <- set_sum_contrasts(h3_data, factors = c("interaction_group", "vis_type"))

h3_model <- lm(
  interaction_composite ~ interaction_group * vis_type + participant_id,
  data = h3_model_data,
  na.action = na.exclude
)

h3_singular <- NA

h3_condition_means <- as.data.frame(
  summary(emmeans(h3_model, ~ interaction_group * vis_type), infer = FALSE)
) %>%
  mutate(
    condition = paste(interaction_group, vis_type)
  )

h3_interaction_means <- as.data.frame(
  summary(emmeans(h3_model, ~ interaction_group), infer = FALSE)
)

h3_vis_means <- as.data.frame(
  summary(emmeans(h3_model, ~ vis_type), infer = FALSE)
)

knitr::kable(
  h3_condition_means %>%
    transmute(
      condition,
      interaction_group,
      vis_type,
      mean = emmean,
      se = SE
    ),
  digits = 4,
  caption = "H3 Estimated Means by Individual Condition (Interaction x Visualization)"
)
```

H3 Estimated Means by Individual Condition (Interaction x Visualization)

condition	interaction_group	vis_type	mean	se
Static PI	Static	PI	3.6256	0.0648
Interactive PI	Interactive	PI	3.8459	0.0639
Buggy PI	Buggy	PI	3.8254	0.0664
Static Ensemble	Static	Ensemble	3.3838	0.0664
Interactive Ensemble	Interactive	Ensemble	3.9451	0.0664
Buggy Ensemble	Buggy	Ensemble	3.7114	0.0654
Static PI + Ensemble	Static	PI + Ensemble	3.3082	0.0654
Interactive PI + Ensemble	Interactive	PI + Ensemble	3.7779	0.0664
Buggy PI + Ensemble	Buggy	PI + Ensemble	3.6679	0.0648

```
knitr::kable(
  h3_interaction_means %>%
    transmute(
      interaction_group,
      mean = emmean,
      se = SE
    ),
  digits = 4,
  caption = "H3 Estimated Means by Interaction Group"
)
```

H3 Estimated Means by Interaction Group

interaction_group	mean	se
Static	3.4392	0.0315
Interactive	3.8563	0.0315
Buggy	3.7349	0.0315

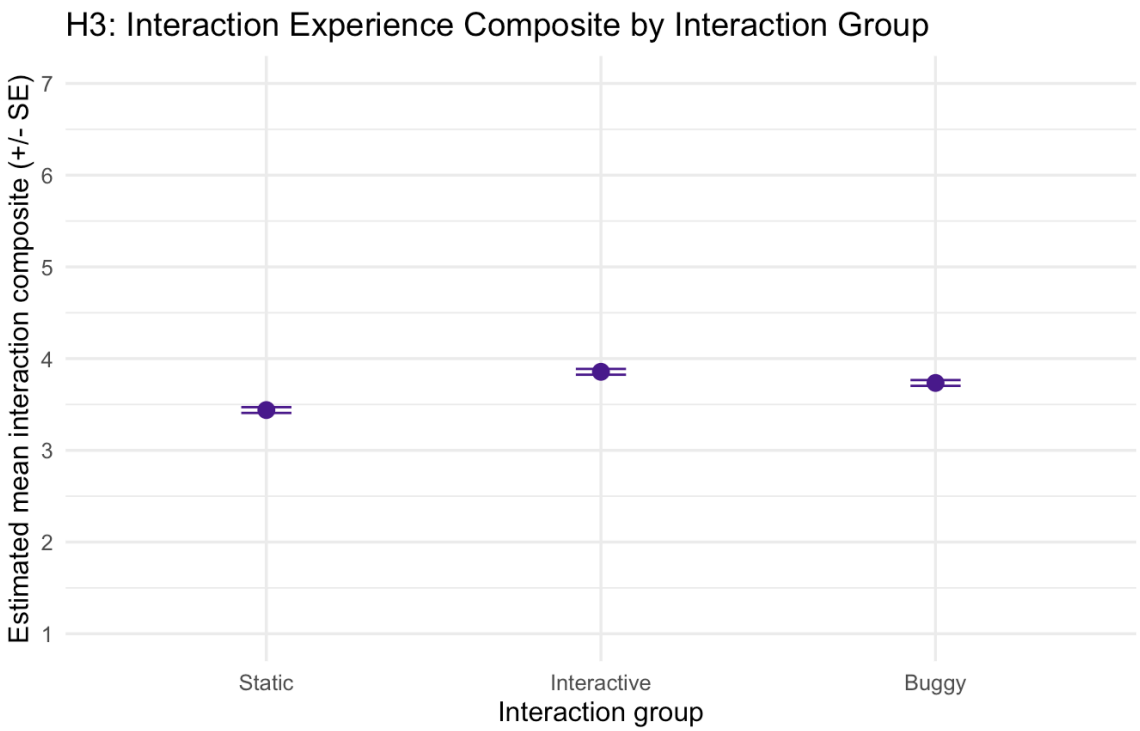
```
knitr::kable(
  h3_vis_means %>%
    transmute(
      vis_type,
      mean = emmean,
      se = SE
    ),
  digits = 4,
  caption = "H3 Estimated Means by Visualization Family"
)
```

H3 Estimated Means by Visualization Family

vis_type	mean	se
PI	3.7656	0.0374
Ensemble	3.6801	0.0382
PI + Ensemble	3.5847	0.0315

```
h3_plot_data <- h3_interaction_means %>%
  mutate(
    ymin = pmax(1, emmean - SE),
    ymax = pmin(7, emmean + SE)
  )

ggplot(h3_plot_data, aes(x = interaction_group, y = emmean)) +
  geom_point(size = 3, color = "#4A148C") +
  geom_errorbar(aes(ymin = ymin, ymax = ymax), width = 0.15, color = "#4A148C") +
  scale_y_continuous(limits = c(1, 7), breaks = 1:7) +
  labs(
    title = "H3: Interaction Experience Composite by Interaction Group",
    x = "Interaction group",
    y = "Estimated mean interaction composite (+/- SE)"
  ) +
  theme_minimal(base_size = 12)
```



Model Execution Checks

```
model_checks <- data.frame(
  hypothesis = c("H1", "H2", "H3"),
  rows_used = c(nrow(h1_data), nrow(h2_data), nrow(h3_data)),
  participants = c(
    dplyr::n_distinct(h1_data$participant_id),
    dplyr::n_distinct(h2_data$participant_id),
    dplyr::n_distinct(h3_data$participant_id)
  ),
  vis_levels_present = c(
    all(c("PI", "Ensemble", "PI + Ensemble") %in% levels(h1_data$vis_type)),
    all(c("PI", "Ensemble", "PI + Ensemble") %in% levels(h2_data$vis_type)),
    all(c("PI", "Ensemble", "PI + Ensemble") %in% levels(h3_data$vis_type))
  ),
  interaction_levels_present = c(
    all(c("Static", "Interactive", "Buggy") %in% levels(h1_data$interaction_group)),
    all(c("Static", "Interactive", "Buggy") %in% levels(h2_data$interaction_group)),
    all(c("Static", "Interactive", "Buggy") %in% levels(h3_data$interaction_group))
  ),
  singular_lmm = c(h1_singular, h2_singular, h3_singular)
)

knitr::kable(model_checks, caption = "Model Execution Validation")
```

Model Execution Validation

hypothesis	rows_used	participants	vis_levels_present	interaction_levels_present	singular_lmm
H1	1005	335	TRUE	TRUE	NA
H2	1005	335	TRUE	TRUE	NA
H3	1005	335	TRUE	TRUE	NA

Additional ANOVA: Visualization Family Within Interactive

Availability Only

```
interactive_vis_data <- h2_data %>%
  filter(interaction_group == "Interactive") %>%
  mutate(
    vis_type = factor(vis_type, levels = c("PI", "Ensemble", "PI + Ensemble"))
  ) %>%
  droplevels()

if (nrow(interactive_vis_data) == 0) {
  stop("No rows found for interaction_group == 'Interactive'.")
}

interactive_vis_model <- lm(trust_composite ~ vis_type, data = interactive_vis_data)
interactive_vis_anova <- anova(interactive_vis_model)

interactive_vis_summary <- interactive_vis_data %>%
  group_by(vis_type) %>%
  summarise(
    n = n(),
    mean = mean(trust_composite, na.rm = TRUE),
    sd = sd(trust_composite, na.rm = TRUE),
    se = sd / sqrt(pmax(n, 1)),
    .groups = "drop"
  )

interactive_vis_anova
```

```
## Analysis of Variance Table
##
## Response: trust_composite
##            Df Sum Sq Mean Sq F value   Pr(>F)
## vis_type    2  14.30  7.1512  5.1143 0.006492 **
## Residuals 332 464.23  1.3983
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
knitr::kable(
  interactive_vis_summary,
  digits = 4,
  caption = "Interactive-only Trust Composite by Visualization Family"
)
```

Interactive-only Trust Composite by Visualization Family

vis_type	n	mean	sd	se
PI	117	4.9872	1.0916	0.1009
Ensemble	109	4.7317	1.1137	0.1067
PI + Ensemble	109	4.4839	1.3335	0.1277